

Exploring Creativity Support for Concept Art Ideation

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ABSTRACT

Creatives often struggle with fixation on a narrow set of ideas. There is potential for co-creative systems to stimulate creatives in new and powerful ways. We sought to lay a foundation for such systems with an exploratory study. We recruited 20 university students and asked them to rapidly draw a series of creature concepts with two technology probes, one of which generates ambiguous stimuli from user strokes. We analyzed the 240 sketches visually and discovered that while most participants were fixated on humanoid forms, those that began sketching with the ambiguous stimuli first were provoked to explore more unusual varieties ($p < 0.01$). We also interviewed participants and used thematic analysis to analyze the data. While some participants resisted the partial loss of control and freedom, most believed the stimuli encouraged more divergent exploration and holistic thinking, and they acknowledged their potential benefit in the earliest stages of the ideation process.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**; • **Applied computing** → **Computer-aided design**.

KEYWORDS

Human-Computer Interaction, Creativity Support, Co-creative System, Drawing, Sketching, Concept Art, User Experience Design

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1 INTRODUCTION

Ambiguity, by definition, is the quality of being open to more than one interpretation. Since multiple interpretations by a single individual or across several individuals can lead to multiple solutions [24], ambiguity has long been considered a useful tool in

the creative process for designers, artists, and other creative professionals. Renowned concept artist and art educator Scott Robertson regularly uses and encourages ambiguous underlay drawings in his approach to concept art [23]. Gaver et al. support ambiguity as being a wonderful resource for design [9].

Sketching is a domain where ambiguity can be utilized readily. In addition to facilitating one's internal thought processes [25] and problem-solving ability [29], sketching can be considered a “dialectic” thinking exercise [11]. Sketches form a dialogue between the brain and the external artifact that is constantly evolving and changing. When the artifact is ambiguous, it holds potential for many different interpretations and new directions. In creative professions like concept art, animation, and game design, creativity and original ideas are highly valued and sought after. Our curiosity in this research centered around the role technology can play in promoting ambiguity and thus creativity in the creative process while sketching. We primarily wanted to understand the effects of *ambiguous stimuli*—which provide the brain with sensory information that contains conflicting evidence for multiple mutually exclusive interpretations [15]—on novice and intermediate creators who would likely benefit the most from assistance from such technology.

We recruited 20 visual arts university students, most of whom are aspiring concept artists, animators, and game designers, and conducted a study that involved sketching twelve alien creature concepts. For half of the concepts the students sketched in a normal manner on a digital tablet, and for the other half they were exposed to a technology probe that generates ambiguous stimuli underlays based on the user's strokes. We make the following contributions in this work:

- A discovery that using ambiguous stimuli in the beginning of the ideation process resulted in less fixation on humanoid forms and provoked more unusual forms. Other effects included cultivating holistic thinking, stimulating exploration, enjoyment, resistance to a loss of control and freedom, and a consensus that the stimuli are most useful in the earliest stages of the creative process.
- Implications for further research in utilizing ambiguous stimuli to augment human creativity and provoke more exploration, along with design implications for related co-creative systems and creativity support tools.

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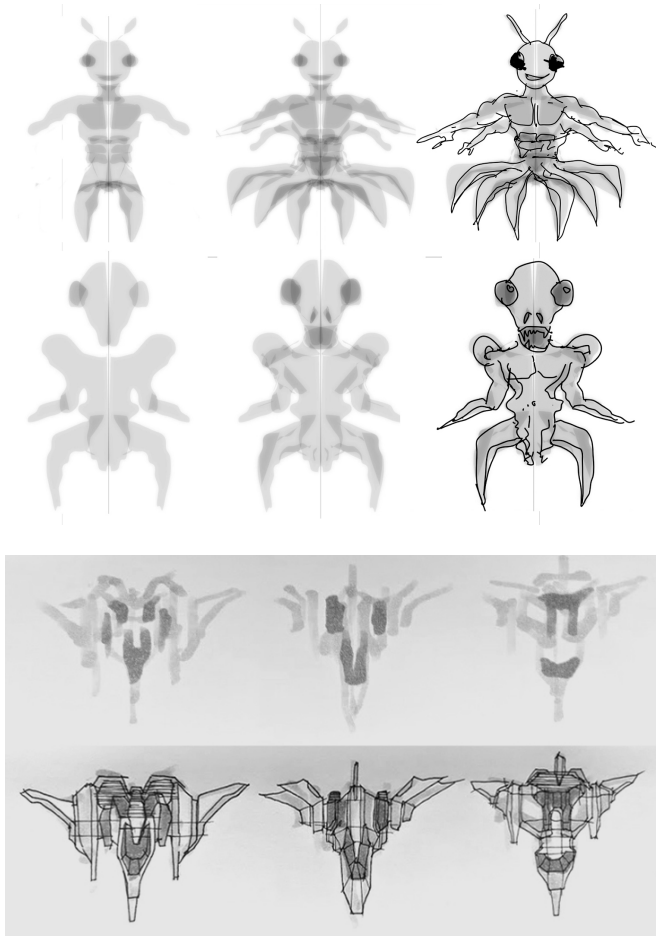


Figure 1: Examples of ambiguous stimuli that aid the user in developing creative concepts.

2 RELATED WORK

The first area of related work focuses on creativity research. Many studies related to human creativity have involved sketching. In particular, the work of Ward’s “structured imagination” studies involved participants sketching alien creatures [27]. He discovered that existing conceptual frameworks could constrain creativity. Many of the participants drew bilaterally symmetric creatures with limbs, sensory organs, etc. even when asked to imagine creatures from a planet very different than Earth. We draw inspiration from both the subject of this study (alien creatures) as well as his approach to coding the drawings visually. In the summative work *Creative Cognition* [7], Finke et al. discuss the importance of “preinventive forms” in the generative stage of the creative process. They found that people are more creative when they generate their own preinventive forms as opposed to building off of pre-existing ones. This is influential in how we designed our prototype to be user-driven and co-creative rather than providing users with ambiguous stimuli to start from.

The second area of related work is concept art techniques. Professional concept artists and art educators like Scott Robertson have popularized techniques for producing concept art that utilize ambiguous stimuli such as light gray marker underlays [16, 23]. These underlays serve as guides for line work, but they are ambiguous and flexible enough to allow for design changes as the artist or designer ideates. This technique is sometimes referred to as *value sketching* as the initial focus is on value and overall form development rather than line work and details [23]. We believe technology can play a role in making these underlays more dynamic and stimulating. Digitally drawn strokes can be abstracted in unique and provocative ways [11]. While this can reduce the freedom and control the user has, it could lead to more divergent thinking and thus more creative concepts.

The third area of related work is computational creativity. Computational co-creative systems are systems that facilitate the co-creation of artifacts between users and computers [4, 8, 13]. Also referred to as Mixed-Initiative Creative Interfaces (MICIs) [28], such systems have been considered in a variety of domains such as public art installations [21], music improvisation [22], poetry writing [3], dancing [20], design ideation [14], and so on. A relevant example of such a system would be the *Drawing Apprentice* from Davis et al [5, 6]. This system responds to user strokes with strokes of its own, resulting in abstract co-created drawings. We draw inspiration from this work, however, we focus more on formalized concept sketching rather than making abstract art. In other words, we focus on a more practical use case of sketching specific concepts or ideas. Another relevant example in this domain is *ShadowDraw* from Lee et al. [19]. This system responds to user strokes by recognizing what the user is trying to draw and providing underlays of that object to assist the user. For example, a user can begin drawing a bicycle, and *ShadowDraw* will generate underlays of bicycle drawings which the user can then trace, helping the user to converge on a stronger bicycle drawing. We draw inspiration from the underlay aspect of this work in particular, but our approach is oriented more towards divergent thinking rather than convergent thinking.

3 TECHNOLOGY PROBES

We built two web-based drawing technology probes, each with a basic set of drawing tools and limited options. We were not focused on providing the users with power and features; rather, we wanted to ensure they could quickly sketch ideas and be focused on those ideas rather than the user interface. One of the tools has the additional capability of allowing the user to co-create ambiguous stimuli with the system. It does this by converting the user’s stroke input into shapes of semi-transparent light gray value and mirroring those shapes. The probe allows for a simple series of strokes to yield fairly complex, layered, and symmetrical ambiguous stimuli which can then serve as an underlay for normal line work. While symmetry could be optional in a more feature-rich co-creative system, we did not make turning off symmetry possible for the study. The symmetry can lead to unexpected and stimulating forms which we considered an important aspect of our exploration on ambiguous stimuli.

4 METHODOLOGY

We conducted a study primarily aimed at answering our research question: *How do these ambiguous stimuli affect an individual's creativity and form ideation?* We chose a within-subjects experiment design where all participants used the ambiguous stimuli probe as well as a basic sketching interface as a control condition. For each participant, we alternated between which condition they were exposed to first, as we were also interested in learning if ambiguous stimuli could have lingering effects on their ideation process. We recruited 20 university students for the study, most of whom were pursuing a degree in Visualization. The details of the demographics and our participants can be found in Appendix A. After answering some demographic questions and self-assessing their ability, participants were prompted with the following task: *An animation studio has recruited you to design an alien creature which will star in their upcoming animated film. Your task is to generate 12 early-stage concepts for this creature.*

Each participant was asked to draw 12 creatures, each having a time limit of four minutes. Drawing inspiration from Shah's *C-Sketch* [24] study, we chose a four-minute time limit because it gave students enough time to fully flesh out a concept. For six of the creatures, the participants used the control probe (basic sketching interface), and for the remaining six they used the ambiguous stimuli probe (experimental condition). For the ambiguous stimuli probe, the four minutes was split between two minutes of creating the ambiguous stimuli underlay and two minutes of normal sketching. This was timed automatically by the system. We alternated between participants being exposed to the control condition first or the experimental condition first as we were interested in observing if that had any lingering effects on their creativity or fixation. Participants were encouraged to think aloud and discuss their creative process as they sketched, including narratives around their creatures, yielding rich qualitative data.

For the sketch data, we saved static PNG images of each creature sketch as well as time-stamped vector stroke data in JSON format to a MySQL database. A total of 240 sketches were collected. We analyzed the sketch data by randomizing the images and coding them visually [2, 17] based on themes and patterns seen in the images. We also took inspiration from previous creativity research in which sketches were coded visually, such as Ward's studies on structured imagination [27]. Our visual analysis resulted in six types of creatures, four "novelty factors", and the emergence of four key findings. We used thematic analysis [1] and the constant comparative method [10] to analyze the qualitative data. In the initial coding phase, we analyzed 362 codes from transcripts and notes gained from the studies. After focused coding and theoretical coding, six categories emerged which are described in the results. Theoretical saturation was reached when data from the last several participants did not add any new categories or dimensions to the existing categories.

5 RESULTS

5.1 Visual Analysis

Our visual analysis resulted in the categories as follows:

- **Creature Types:** *Humanoid*: primarily bipedal, with 2 arms and 2 legs; *Abstract*: amorphous / blob-like, or do not fit into any other category; *Insectoid*: resembles insects including butterflies, moths, and so on; *Mammalian*: resembles terrestrial mammals such as dogs, cats, bears, and so on; *Aquatic*: resembles fish, squids, octopi, jellyfish, and so on; and *Inorganic*: appears to be robotic or an inanimate object.
- **Novelty Factors:** *Proportions*: unusual size relationships such as head to body ratio; *Limbs / Appendages*: unusual number or kinds of limbs, appendages, and so on; *Sensory Organs*: unusual number or kinds of eyes, mouths, ears, and so on; and *Clothing / Accessories*: external items such as robes, capes, hats, or weapons.

Note that creature types alone were not enough to sufficiently measure the creativity or novelty of the creature concepts, so novelty factors were an additional metric we introduced to quantify the unique features of the creatures that the participants explored.

Overall fixation on humanoid forms. 43.8% of all creatures sketched were categorized as humanoid, suggesting a fixation on humanoid forms across most participants. This is consistent with similar prior research and is very reminiscent of Ward's "structured imagination" work in which most creatures had human-like traits like limbs and sensory organs [27]. Participants may have been attempting to anthropomorphize their creatures and make them relatable due to the prompt. This can be observed in Figure 2.

Ambiguous stimuli first caused less overall fixation on humanoid forms. The participants who were exposed to the ambiguous stimuli first had had 41% less humanoid forms, 42% more abstract forms, and nearly four times as many inorganic forms, e.g., robots and inanimate objects (see Figure 3). We acknowledge that one participant P12 was clearly fixated on robots. A Pearson's chi-squared test revealed that the effect of beginning with the ambiguous stimuli first was statistically significant ($p < 0.01$) in influencing different creature varieties.

Ambiguous stimuli provoked particular forms more than others. When isolating the sketches from the ambiguous stimuli condition, we found that particular forms were provoked. For example, there were significantly more (12 to 1) insectoid forms. Meanwhile, the control condition resulted in much more inorganic forms such as robots and inanimate objects, as well as more aquatic forms like octopi and squids. This finding is very consistent with the qualitative data and we will touch on it again in Section 5.2. A Pearson's chi-squared test revealed that there was statistical significance ($p < 0.01$) in the effect of ambiguous stimuli on creature varieties.

Ambiguous stimuli don't impede exploration of novelties. The stimuli resulted in the exploration of slightly more novelties including proportions, limbs, and sensory organs, but less of clothing and accessories. A Pearson's chi-squared test revealed that there was statistical significance ($p < 0.02$) in the effect of ambiguous stimuli on exploring more novel features. We recognize that the coding of novelties is highly subjective, so the key finding we are claiming is that ambiguous stimuli *do not impede* an individual's creativity or inclination for exploring novel features. The average

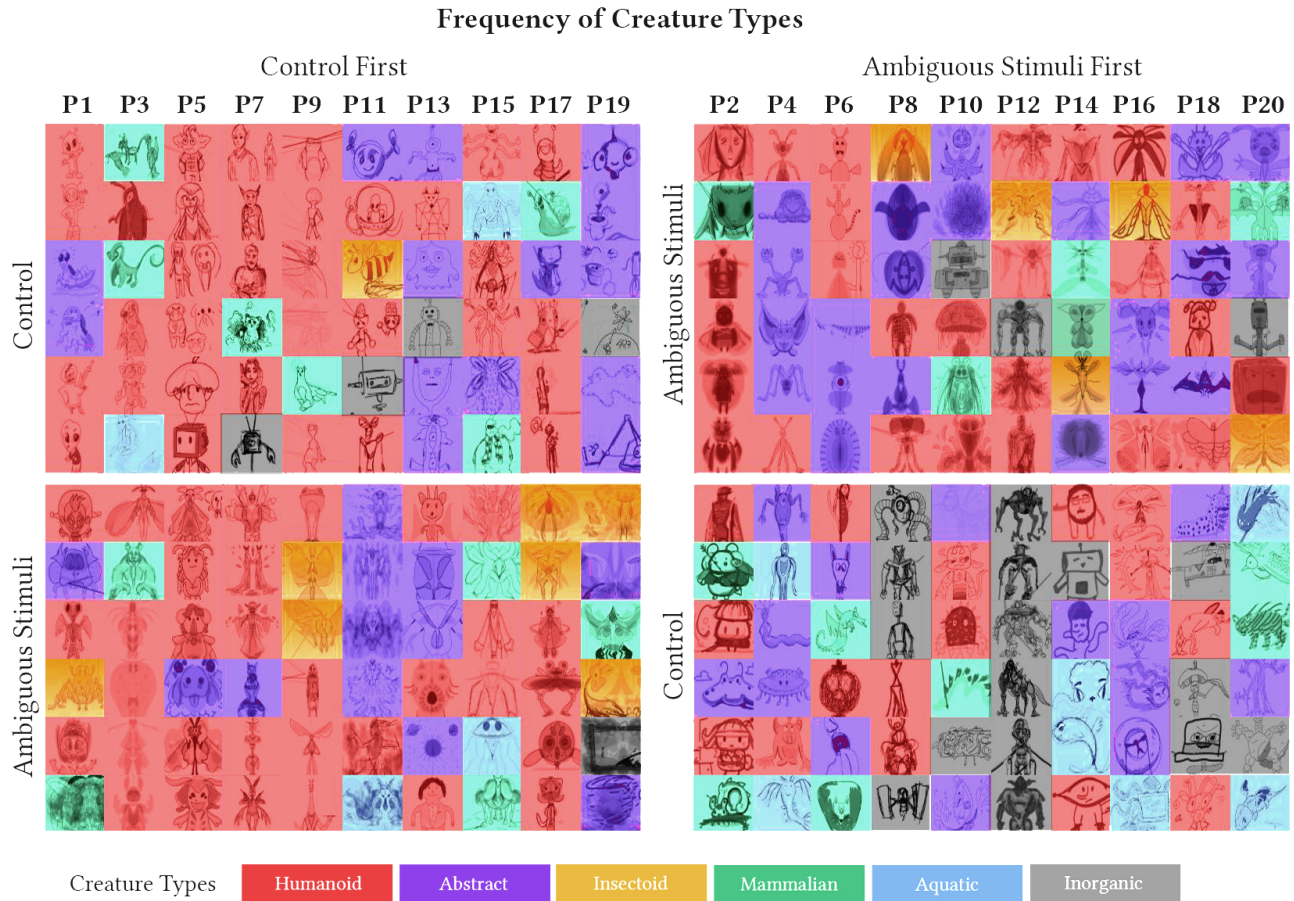


Figure 2: A heatmap visualization of creature types across all participants and both conditions. Note that participants who were exposed to the ambiguous stimuli condition first produced significantly less humanoid forms and produced more abstract and inorganic forms.

participant explored around two novelties (1.98 ± 0.82) per creature concept regardless of condition.

5.2 Qualitative Analysis

Holistic Thinking Cultivated. The ambiguous stimuli seemed to cultivate many participants' focus on the "big-picture" and overall form rather than being caught up in details. This was likely due to the combination of symmetry and value allowing for a full form to emerge very quickly. Some participants expressed that they normally begin by drawing silhouettes, suggesting that the ambiguous stimuli meshed well with their creative process.

"I do like how it made me think in generalities"—P9

"It was forcing me to figure out the shape of the character before the details"—P4

"You can get stuck in the details so it prevents you from that"—P3

Exploration was Stimulated. Many of the participants found that the stimuli cultivated more exploration and led to less fixation on what they normally are accustomed to drawing. Some participants felt that it loosened them up and expressed that it caused them to approach the sketching with less of a preconceived idea. This is consistent with the finding that the participants who were exposed to the ambiguous stimuli first had less fixation on humanoid forms despite the overall trend of humanoid forms across both conditions. It is also consistent with the finding that the ambiguous stimuli condition resulted in different forms and slightly more exploration of novelties.

"It's freeing me up a bit"—P3

"The really expressive shapes keep your mind going. It makes you think of something else"—P4

"After you get used to using it, it gives you an interesting way to think of ideas"—P10

Interesting and Enjoyable. Most of the participants found it interesting and enjoyable to generate the ambiguous stimuli. Most

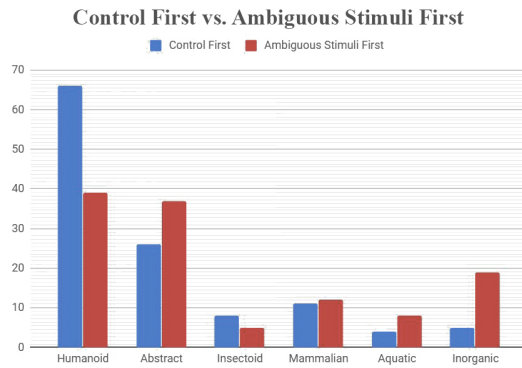


Figure 3: Participants who were exposed to the ambiguous stimuli condition first produced significantly less humanoid forms and instead produced more abstract or inorganic forms.

of them had never used symmetry tools or sketched in such a way before. Some found it relaxing while others found it humorous or pleasantly surprising.

“It’s weird, but not in a bad way”—P9

“It feels very satisfying. It’s super fun to make creature on this”—P2

“This is cool, I’ve never tried drawing like this before”—P7

Particular Forms Provoked. Many of the participants expressed that the stimuli pushed them towards certain specific forms. The most common themes were “insects” and “bugs”, including “butterflies” and “moths”. Some participants also mentioned clothing including “robes” and “tuxedos”. A few participants expressed that the stimuli encouraged “organic” forms as opposed to more angular and inorganic forms. This was not seen as a negative by most participants, although some did feel constrained by it. This is very consistent with the finding from our visual analysis discussed earlier. The ambiguous stimuli did result in significantly more insectoid forms and significantly less inorganic forms such as robots and inanimate objects. This may have been influenced by the particular way we designed the ambiguous stimuli probe.

“It makes me think of insect shapes”—P3

“It’s definitely easier to think of organic forms”—P5

“I felt constrained by bugs”—P9

Loss of Control and Freedom. Some participants were initially jarred by the loss of control in their creative process. The ambiguous stimuli forced symmetry and areas of values which could sometimes be surprising and not what was originally intended. Many expressed that they prefer asymmetry when sketching for a variety of reasons.

“This is not the typical way I draw”—P11

“I definitely feel like the symmetry is constraining me”—P20

“Sometimes you need asymmetry for a character to stand out”—P1

Benefit to Creative Process. Still, most participants acknowledged they could see the benefit of giving up some power and freedom for what the ambiguous stimuli allow for. Participants acknowledged their usefulness in speed, forcing reconsideration, and adding beneficial “chaos” to the creative process. There was also a nearly universal consensus from participants that the ambiguous stimuli would be useful in the earliest stages of ideation but not in any other part of their creative process. This includes “thumbnailing” and the initial effort of generating a variety of ideas from which to later narrow down the best ones. This is consistent with our intention for ambiguous stimuli.

“This would help with production speed. I feel like I can make a lot of silhouettes this way”—P2

“It kind of forces chaos in to your design work flow”—P12

“I’d use it only for thumbnailing and early concepts for characters”—P9

6 DISCUSSION

We acknowledge that this study focused only on university students within one major at one university. The study is also focused on one particular subject (creatures). The claims made on the effects of creativity are therefore preliminary, and should be validated and expanded upon in further research.

Furthering Creativity Research. The results showed that being exposed to the ambiguous stimuli first resulted in less fixation on humanoid forms in subsequent sketches, however the participants were overall still largely fixated on these forms. This is consistent with previous creativity research which suggests that humans rely on categorical knowledge when ideating and that creativity is little different than normal cognition [7, 27]. An interesting further inquiry is how can these stimuli further push people towards less expected results and help them overcome the limitations of “structured imagination”.

Balance Between User, System, and Control. These results lead to an interesting question: How much control should the user and the system each have when generating ambiguous stimuli? Some participants resisted the forced symmetry aspect of the ambiguous stimuli. Allowing them the flexibility to turn this feature off and on is simple to implement and would give them more control and freedom, however it would push the system more towards a standard sketching program. Adding adjustable parameters may give users more flexibility and power and make the ambiguous stimuli more useful and provocative. We purposely avoided making the system in this study too sophisticated and feature-rich, as we wanted to isolate the ambiguous stimuli as a variable. Perhaps allowing the system to automatically adapt to a users sketching style or preferences may be a promising future direction [18]. These “adaptable interfaces” are a promising way of using machine learning to automatically personalize the experience to the user.

Benefits at Various Skill Levels. We believe ambiguous stimuli can be helpful to individuals with a variety of skill levels. We chose to conduct the study with students in creative majors because we sought feedback from people who had a stake in creativity and would benefit the most from co-creative tools. The participant pool was more intermediate in terms of drawing ability on average.

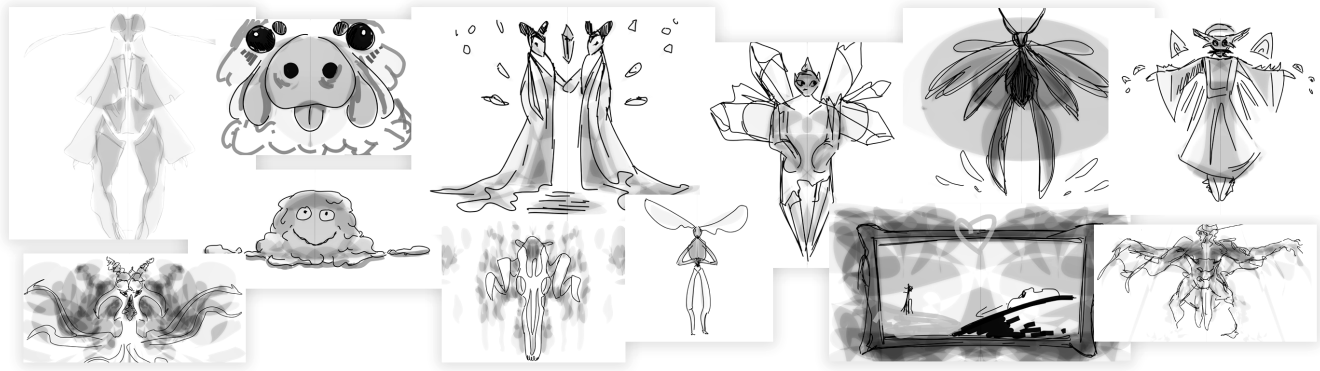


Figure 4: Examples of creature sketches generated by participants with the ambiguous stimuli probe.

However, we believe there is also potential for ambiguous stimuli to help absolute novices overcome a phenomenon described by many drawing instructors and their students as “Fear of the Blank Page” [12]. By quickly gaining momentum through generating ambiguous stimuli, novices could potentially improve their creative self-efficacy [26].

7 CONCLUSION

In this work, we explored the effects of ambiguous stimuli on the ideation process of aspiring creatives. We discovered that while participants were largely fixated on humanoid forms, sketching with the stimuli first seemed to lessen that fixation. We also found that the stimuli influenced sketching particular forms over others, but did not impede individual approaches to creativity and the exploration of novel features. Most participants seemed to enjoy the stimuli, saw benefit in their potential to influence their creative process, and considered them to be most valuable in the earliest stages of ideation. Some participants resisted a partial loss of control and freedom in their creative process, suggesting a better balance between the user and system is important. We find the lessons learned valuable for related research in the domains of co-creative systems, creativity support tools, and creativity research in general. This work serves as a strong step towards augmenting human creativity in the domain of concept art with co-creative systems.

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A PARTICIPANTS

These students are working towards careers in concept art, animation, game design, and other fields related to entertainment. We chose from this pool because we wanted participants who had a stake in creativity and who could potentially benefit the most from creativity support for form ideation. We expected more rich sketch data and qualitative data from these participants.

The participants included 10 males and 10 females at the undergraduate and graduate levels ranging from 19 to 32 years old. In general, participants ranged from novice to intermediate with respect to their confidence in their drawing ability and creativity. Their average self-rating for drawing ability was **3.55 (± 1)** out of 5 while their average self-rating for creativity with respect to drawing was **3.85 (± 0.59)** out of 5.